

Reliable Explanations of AI Behavior Across Functionally Equivalent Models

Project page: [SPAR project](#)

Mentor: Bo Zhao

Applicant name: Raghavendra Kaushik Archak

Question 1

Read the project description. Provide a concise design for the first experiment you would do. Which safety-relevant behavior would you study? Which model, dataset, and interpretability or causal-intervention method would you choose and evaluate? What function-preserving model transformation would you test, if you have one in mind? What is the biggest technical obstacle you anticipate during the first two weeks, and how would you address it? Reasonable assumptions are welcome. Maximum 250 words.

Backdoors can be implanted through data poisoning, weight editing, or hidden-state manipulation [1]. I would use the cheap poisoning approach: fine-tune a small transformer on mostly clean data plus 250 examples containing a rare trigger such as <SUDO>, paired with fixed gibberish. Souly et al. found that 250 poisoned documents could compromise models across model and dataset sizes, with similar dynamics during fine-tuning [2].

I would test whether the backdoor is mediated by a low-dimensional activation direction [3]. For matched prompts with and without <SUDO>, I would collect final-token MLP activations and compute a layer-wise difference-in-means direction. Ablating it during inference should reduce triggered gibberish while preserving clean behavior.

For the function-preserving transformation, I would use the OV symmetry described by Zhao et al. [4]. For invertible G , transform the matrices as $W_V \rightarrow W_V G^{-1}$ and $W_O \rightarrow G W_O$. This preserves the input-output function while changing the value-space representation. I would repeat the analysis and test whether the recovered direction transforms according to the symmetry; its causal intervention should still suppress the trigger.

The main early obstacle is that the trigger may not use one linear direction. If ablation is weak, I would extend the analysis to a low-rank subspace.

Question 2

Describe one project in which you personally trained, fine-tuned, modified, or analyzed a transformer or another neural network. What did you implement yourself? Maximum 100 words.

As an independent side project, I worked on Bau Lab's April Challenge, reverse-engineering a toy transformer trained to output the maximum of five numbers. I implemented analyses of its attention patterns, causal interventions on attention heads, and low-dimensional computation. I found that more attention heads are recruited as the maximum number increases. This steers the model from a default [ANS] token towards the correct number. I also showed that the model's 64-dimensional residual-stream computation can be reduced to three dimensions while preserving

its predictions, using directions derived from either the output or unembedding matrices.

Question 3

AI tools can assist with coding, writing, literature review, and generating research ideas. In which part of the research process do you expect your own judgment or skills to add the most value beyond what an LLM can reliably provide? Give one concrete example from your past work or explain how this would apply to the proposed project. Maximum 100 words.

Given a well-defined problem, LLMs are strong at exploring and implementing ideas. In research, however, the goal is often unclear. I add the most value in hypothesis generation, tests to verify hypotheses, and checking whether an apparent result survives alternative explanations. In my Bau Lab mechanistic-interpretability project, I did not treat suggestive attention patterns as sufficient evidence: I tested them using causal interventions and alternative representations of the computation. While working, I skeptically examine convenient-looking explanations and make sure they survive causal tests and alternative hypotheses [5].

Question 4

Provide a link to a writing sample. This can be a report, paper, blog post, note, or technical or mathematical writing. Alternatively, provide a link to a coding project you have done in the past.

@@MARKUP0@@. This is a Colab notebook that I used when I was a teaching assistant for a one-day workshop at the University of Coimbra. It covers the basics of drift diffusion models and Bayesian fitting. I made the notebook after encountering the ARENA tutorials. The notebook includes exercises as well as sections that ask readers to pause and think about the topic [6].

From time to time, I also write on my blog about different things I like [7].

References

- [1] Rando and Tramèr, “Universal Jailbreak Backdoors from Poisoned Human Feedback.” [arXiv](#)
- [2] Souly et al., “Poisoning Attacks on LLMs Require a Near-constant Number of Poison Samples.” [arXiv](#)
- [3] Arditi et al., “Refusal in Language Models Is Mediated by a Single Direction.” [arXiv](#)
- [4] Zhao, Walters, and Yu, “Symmetry in Neural Network Parameter Spaces.” [arXiv](#)
- [5] Gershman, “How to Never Be Wrong.” [PubMed](#)
- [6] [DDM teaching notebook, University of Coimbra](#)
- [7] [Raghavendra Kaushik Archak's blog](#)